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Honeycomb extrusion dies and methods of using and making same

Abstract

A honeycomb extrusion die (120) with improved wear properties. Extrusion die has a die body (121) with inlet (122) and exit (123) faces, feedholes (124) with feedhole entrances (124A) and outlets (124B), and a plurality of die pins (126) having side surfaces (128) configured to define a matrix of intersecting slots (130). At least some of the intersecting slots and die pins define a slot structure with divots (132) formed in the side surfaces of the die pins between the feedholes and the exit face, entrance slot portions between the feedhole outlets and the divots, the entrance slot portions having an entrance slot width $W_{sub.A}$, and exit slot portions between the divots and the exit face, the exit slot portions having an exit slot width $W_{sub.B}$, wherein $W_{sub.A} > W_{sub.B}$ over an entire slot length. Methods of manufacturing honeycomb structures using the honeycomb extrusion dies and of fabricating the extrusion dies are provided as are other aspects.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATION (1) This is a national stage application under 35 U.S.C. § 371 of International Application No. PCT/US2022/029735, filed on May 18, 2022, which claims the benefit of priority under 35 U.S.C. § 119 of U.S. Provisional Application Ser. No. 63/193,764 filed on May 27, 2021, the content of which is relied upon and incorporated herein by reference their entireties.

FIELD

(1) Embodiments of the disclosure relate to honeycomb extrusion dies, and in particular to honeycomb extrusion dies comprising slots with divots extending into side surfaces of the slots and methods of using such honeycomb extrusion dies to manufacture honeycomb structures.

BACKGROUND

(2) Honeycomb extrusion dies can be used to extrude honeycomb structures from ceramic-forming batch materials, which after firing, may produce porous honeycomb structures that can be utilized in fluid treatment systems, such as catalytic converters and particulate filters, for example.

SUMMARY

(3) An example embodiment of the disclosure provides a honeycomb extrusion die. The honeycomb extrusion die comprises a die body comprising an inlet face and an exit face, the die body having feedholes with feedhole entrances at the inlet face and feedhole outlets, and a plurality of die pins extending to the exit face, the plurality of die pins comprising side surfaces configured to define a matrix of intersecting slots, at least some of the matrix of intersecting slots and the plurality of die pins further comprising: divots formed in the side surfaces of the die pins between the feedholes and the exit face, entrance slot portions of the matrix of intersecting slots between the feedhole outlets and the divots, the entrance slot portions having an entrance slot width $W_{sub.A}$, and exit slot portions of the matrix of intersecting slots between the divots and the exit face, the exit slot portions having an exit slot width $W_{sub.B}$, wherein $W_{sub.A} > W_{sub.B}$ over an entire length from the feedhole outlets to the divots.

(4) Another example embodiment of the disclosure provides a honeycomb extrusion die. The honeycomb extrusion die comprises a die body comprising an inlet face and an exit face, the die body having feedholes with feedhole entrances at the inlet face and feedhole outlets, and a plurality of die pins extending to the exit face, the plurality of die pins comprising side surfaces configured to define a matrix of intersecting slots, at least some of the matrix of intersecting slots and the plurality of die pins further comprising: divots formed in the side surfaces of the die pins between the feedholes and the exit face, entrance slot portions of the matrix of intersecting slots between the feedhole outlets and the divots, the entrance slot portions having an entrance slot width $W_{sub.A}$, and exit slot portions of the matrix of intersecting slots between the divots and the exit face, the exit slot portions having an exit slot width $W_{sub.B}$, wherein $W_{sub.A} > W_{sub.B}$ over an entire length from the feed hole exits to the divots, and wherein the at least some of the matrix of intersecting slots have SWCR of greater than or equal to 1.5, wherein SWCR is a slot width contraction ratio defined as $SWCR = W_{sub.A} / W_{sub.B}$, and wherein the at least some of the entrance slot portions have an entrance length $L_{sub.A}$ as measured between the feed hole outlets and the divots, and at least some of the exit slot portions have an exit length $L_{sub.B}$ as measured between the divots and the exit face, and wherein $L_{sub.A} > L_{sub.B}$.

(5) Another example embodiment of disclosure provides a method of manufacturing a honeycomb structure. The method comprises providing a honeycomb extrusion die having a die body comprising: an inlet face and an exit face, the die body having feedholes with feedhole entrances at

the inlet face and feedhole outlets, and a plurality of die pins extending to the exit face, the plurality of die pins comprising side surfaces configured to define a matrix of intersecting slots, at least some of the matrix of intersecting slots and the plurality of die pins further comprising: divots formed in the side surfaces of the die pins between the feedholes and the exit face, entrance slot portions of the matrix of intersecting slots between the feedhole outlets and the divots, the entrance slot portions having an entrance slot width $W_{sub.A}$, and exit slot portions of the matrix of intersecting slots between the divots and the exit face, the exit slot portions having an exit slot width $W_{sub.B}$, wherein $W_{sub.A} > W_{sub.B}$ over an entire length from the feedhole outlets to the divots; and extruding a batch material through the feedholes and matrix of intersecting slots. A position of maximum die wear from the extrusion is located at the exit slot portions of the matrix of intersecting slots, regardless of the rheology of the batch material.

(6) Additional features of the disclosure will be set forth in the description which follows, and in part will be apparent from the description, or may be learned by practice of the disclosure. It is to be understood that both the foregoing general description and the following detailed description are explanatory and are intended to provide further explanation of the disclosure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The accompanying drawings, which are included to provide a further understanding of the disclosure are incorporated in and constitute a part of this specification, illustrate embodiments of the disclosure and together with the description serve to explain the disclosure. The drawings are not necessarily drawn to scale.

(2) FIG. 1 schematically illustrates a cross-sectional side view of a representative portion a honeycomb extrusion die including a plurality of slots including slot entrance portions, slot exit portions, and intermediate divots wherein a width $W_{sub.A}$ of the slot entrance portions is greater than a width $W_{sub.B}$ of the slot exit portions according one or more embodiments of the disclosure.

(3) FIG. 2 schematically illustrates a partially cross-sectioned, side view diagram of an extruder (e.g., twin-screw extruder) including an extrusion die located at a discharge end according to embodiments of the disclosure.

(4) FIG. 3 schematically illustrates a perspective, partial view diagram of an extruder front end and a green honeycomb structure being extruded from a honeycomb extrusion die according to embodiments of the disclosure.

(5) FIG. 4 schematically illustrates a perspective view of a representative honeycomb body comprising a honeycomb structure manufactures by extrusion through the honeycomb extrusion die according to embodiments of the disclosure.

(6) FIG. 5 schematically illustrates an enlarged, cross-sectioned, front plan view of a representative portion of a honeycomb extrusion die illustrating locations of a plurality of feedholes in fluid communication with the plurality of slots and with a representative die pin configuration including a cutout portion illustrating configurations of slot entrance portions, slot exit portions, and intermediate divots according to embodiments of the disclosure.

(7) FIG. 6 schematically illustrates an enlarged cross-sectional side view of a representative portion of a honeycomb extrusion die showing a slot configuration including slot entrance portions, slot exit portions, and intermediate divots according to embodiments of the disclosure.

(8) FIG. 7A schematically illustrates an enlarged cross-sectional side view of a representative portion of an alternative embodiment of a honeycomb extrusion die including tapered slot entrance portions according to some embodiments of the disclosure.

(9) FIG. 7B schematically illustrates an enlarged partial cross-sectional side view of a

representative portion of an alternative embodiment of a honeycomb extrusion die including square divot shapes according to some embodiments of the disclosure.

(10) FIG. 8 schematically illustrates flowchart describing a method of manufacturing a honeycomb structure using a honeycomb extrusion die containing slot entrance portions, slot exit portions, and intermediate divots according to embodiments of the disclosure according to embodiments of the disclosure.

DETAILED DESCRIPTION

(11) Honeycomb bodies are used in many different applications. For example, exhaust after-treatment systems of exhaust gas from internal combustion engines may use one or more catalysts supported on high-surface area substrates (e.g., ceramic honeycomb bodies) to reduce exhaust pollutants such as CO, HC, NO_x, SO_x, for example. Similarly, high porosity honeycomb bodies may be end plugged to for use as wall-flow honeycomb filters. Porous ceramic bodies useful as catalyzed flow-through honeycomb substrates and wall-flow honeycomb filters can be manufactured utilizing the honeycomb extrusion dies according to the embodiments described herein.

(12) In particular, a honeycomb structure of a honeycomb body can be formed by extruding from a honeycomb extrusion die, a ceramic-forming batch mixture, comprising ceramic-forming materials that may comprise ceramics or ceramic precursors, or both, an organic binder(s), a liquid vehicle, an optional pore former, and possibly other processing aids, lubricants, or sintering aids. After green honeycomb formation (e.g., via extrusion through the honeycomb extrusion die), the green honeycomb body can be dried and reacted and/or sintered into a porous ceramic honeycomb body. The porous ceramic honeycomb body can have open and interconnected porous ceramic honeycomb macro- and microstructure suitable for exhaust after-treatment or other fluid treatment purposes. The ceramic material may be, for example, cordierite, aluminum titanate, alumina, mullite, silicon carbide, silicon nitride, and the like, and combinations thereof. Other suitable extrudable batch materials may be used.

(13) The extrusion can be performed using any suitable extruder, such as a hydraulic ram extrusion press, a two stage de-airing single auger extruder, a twin-screw extruder, and the like; each with a honeycomb extrusion die of an embodiment as described herein attached to the discharge end thereof, such as shown in FIGS. 2 and 3, for example.

(14) As should be recognized, extrusion of such ceramic-forming batch mixtures through the honeycomb extrusion die subjects the honeycomb extrusion die to severe wear given the abrasive nature of the batch mixture being extruded. The amount of wear may be dependent on the properties and components of batch mixture and its corresponding liquid content. For example, more coarse batch mixtures (e.g., larger particle sizes) may cause more aggressive wear, and likewise dryer batches (containing less liquid) may cause more wear. Such honeycomb extrusion dies have tolerances and/or features that are desired be maintained during the useful life of the extrusion die. However at some point in the life of the extrusion die, these tolerances and/or features may become degraded to the point that desired honeycomb dimensions (e.g., honeycomb wall thickness) or features in the extruded bodies may no longer be obtained. Although honeycomb extrusion dies can be conventionally recoated with a suitable wear coating to extend their useful life, recoating can be expensive, and the extent of recoating is generally limited, as at some point the honeycomb extrusion die can no longer be used (e.g., once it has been excessively degraded from both use and the coating operations). Thus, extrusion die designs that assist in enabling improved wear characteristics and attributes may be effective in increasing the useful life of the honeycomb extrusion die. This can have significant impacts to the cost of manufacturing honeycomb bodies and to the quality of the resulting honeycomb bodies.

(15) One goal in producing some honeycomb bodies is to minimize wall thickness variability, for example, as both the variance in the dimensions of the walls of a single honeycomb body, and also as variance in the dimensions of multiple honeycomb bodies made from the same die. For example,

as an extrusion die wears, the width of the slots used to form the honeycomb walls can widen and become increasingly less uniform across the die. This variability, such as in the form of a slot width standard deviation, is one metric that may help to determine if a particular honeycomb extrusion die is still capable of producing a honeycomb body within acceptable dimensional tolerances or limits. In some cases, a honeycomb extrusion die may be rejected after extrusion of a certain linear footage of extruded green ware due to increases in slot width variability correlated to this linear footage. In other cases, the honeycomb extrusion die may be inspected by image-analysis software. (16) In addition to altering the dimensions, or the variability in the dimensions, of the extruded honeycomb, wearing of a die pin root (at the location where the feedholes intersect with the slots) may significantly reduce the cross-sectional area of the attachment of the die pin to the die body, and thus reduce its area moment of inertia. This reduced area moment of inertia may lead to higher stresses when a die pin is subject to any bending moment, such as during extrusion, handling, and/or during die recoating. Particularly, non-uniform wear at the die pin root may lead to die pin movement during recoating due to the shape asymmetry of the coating and the die pin. Additionally, this asymmetry coupled with the difference in coefficient of thermal expansion (CTE) between the base material and wear coating, may cause further die pin deformation. This may result in even greater slot width variability.

(17) Another contributor to slot width variability can be wear of the die pin sides at the exit of the slots, at which location the width of the extruded honeycomb walls are determined. For abrasive batches with high wall drag, the wear of the die may occur at the highest rate at the die pin root at the intersection of feedhole and slot. In accordance with the above, wear at the die root can lead to a premature ending of the die's life, e.g., prior to its expected linear footage. While wear at the exits of the slots may eventually lead to unacceptably high slot width variability, extrusions processes are generally less sensitive to wear at the slot exits than to wear at the pin roots.

(18) Although die wear is a generally unavoidable consequence of extrusion of abrasive batch materials, one advantage of the honeycomb extrusion die embodiments described herein is that the wear is reduced and the location of the wear is controlled so that it is in an area of the die that is less susceptible to corresponding dimensional variability in the extruded honeycomb bodies. That is, to improve honeycomb extrusion die life, in accordance with embodiments described herein, the die is arranged so that the highest rate of wear occurs at the slot exits, and this is accomplished regardless of batch mixture used, i.e., independent of the rheological characteristics of the batch mixture. For example, the inventors have discovered that less wall thickness variability occurs in the extruded honeycomb, and honeycomb extrusion die life can be extended, if the peak batch slip velocity (representing wear) at the intersection between the feedholes and slots is less than the batch slip velocity at the exit of the slots.

(19) In addition to increased die life, the embodiments described herein also allow for lower die pressure due to less wear at the pin root enabling dies having comparatively wider slot widths at the intersections between the feedholes and the slots. In turn, lower die pressures may assist in enabling dies having longer feedhole lengths, which may reduce die crowning (and thus be particularly useful for larger diameter dies, which may undergo significant crowning during use). As described herein, the disclosed embodiments assist in providing a single design that yields consistent performance (e.g., location of highest wear) for all honeycomb extrusion dies regardless of batch rheology, e.g., regardless of whether the batch material is a so-called low wall-drag batch or a high wall-drag batch.

(20) Referring now to FIG. 1, a honeycomb extrusion die **120** is illustrated that comprises a die body **121**. The die body **121** comprises an inlet face **122** and an exit face **123**, which are axially opposite from each other. For example, in conjunction with FIGS. 2-3 described in more detail below, the inlet face **122** receives batch material, such as the batch material **216**, via the action of an extruder, such as an extruder **200**, and the exit face **123** discharges the batch material **216** reformed as a honeycomb structure **201**.

(21) Referring back to FIG. 1, the die body **121** comprises feedholes **124** extending in the die body **121** with feedhole entrances **124A** at the inlet face **122** and feedhole outlets **124B** within the body **121**. A plurality of die pins **126** extend from the die body **121** to form the exit face **123**. The plurality of die pins **126** comprise side surfaces **128** (a few labeled) that define a matrix of intersecting slots **130**. The matrix of intersecting slots **130** corresponds to the extruded honeycomb structure **201**, as the walls of the honeycomb structure **201** are formed from the batch material **216** as the batch material **216** is discharged from the slots **130** at the exit face **123**.

(22) At least some of the slots **130** in the matrix of intersecting slots (such as all of the slots **130**) comprise divots **132** formed in the side surfaces **128** of the die pins **126** at positions located between the outlets **124B** of the feedholes **124** and the exit face **123**. Accordingly, these intersecting slots **130** comprise entrance slot portions **135** of the matrix of intersecting slots **130** extending from intersections of the slots with the feedhole exits **124B** to the divots **132**. The entrance slot portions **135** have an entrance slot width $W_{sub.A}$ measured transverse to the slot **130** as best shown in FIGS. 1 and 6. The intersecting slots **130** further comprise exit slot portions **136** located extending between the divots **132** and the exit face **123**. The exit slot portions **136** have an exit slot width $W_{sub.B}$, measured transverse to the slot **130** (perpendicular to the axial direction) as also shown in FIGS. 1 and 6. Since the batch material is discharged from the extrusion die **120** through the slots **130**, the exit slot width $W_{sub.B}$ of the exit slot portions **136** corresponds to the thickness of the walls **312** of the resulting honeycomb structure **201** and honeycomb body **203**.

(23) According to embodiments described herein, in order to set the highest wear area in the honeycomb extrusion die **120** to occur at the location of the exit slot portions **136**, the configuration of the honeycomb extrusion die is made in some embodiments such that $W_{sub.A} > W_{sub.B}$ over an entire length of the slots **130**. In other words, the entrance slot width $W_{sub.A}$ for any location along an entrance length $L_{sub.A}$ of the entrance slot portions **135** (from the intersections with the feed hole exits **124B** to the divots **132**, as shown in FIG. 6) is greater than the exit slot width $W_{sub.B}$ for all locations along an exit length $L_{sub.B}$ of the exit slot portions **136** (from the divots **132** to the exit face **123**, as shown in FIG. 6).

(24) Reference is now made to FIG. 2, which shows a cross-sectioned side view of a schematic example embodiment of an extruder **200** (e.g., a continuous twin-screw extruder). The extruder **200** includes the honeycomb extrusion die **120** according to embodiments of this disclosure mounted at a discharge end at a downstream side **210** of the extruder **200**. The extruder **200** comprises a barrel **202** comprising a chamber **204** formed therein. The barrel **202** can be monolithic or it can be formed from a plurality of barrel segments connected successively in the longitudinal (e.g., axial) direction **240**, which corresponds to the extrusion direction. The chamber **204** extends through the barrel **202** in the longitudinal direction **240** between an upstream side **208** and the downstream side **210**. At, or proximate to, the upstream side **208** of the barrel **202**, a material supply port **214** can be provided, which can comprise a hopper or other suitable material supply structure, for supplying a batch material **216** to the extruder **200**. The honeycomb extrusion die **120** is provided at, and coupled to, the downstream side **210** of the barrel **202** for extruding the batch material **216** into a desired shape, such as honeycomb structure **201** or the like. Thus, the honeycomb structure **201** is illustrated in FIG. 2 as a green extrudate that is in the process of being extruded through the die **120**. The honeycomb extrusion die **120** can be preceded by other extrusion components (not shown), such as one or more screen, orifices, homogenizers, flow control or bow control devices, or the like to facilitate the formation of desirable flow characteristics, such as a steady, uniform, and/or plug-type flow front before the batch material **216** reaches the honeycomb extrusion die **120**. FIG. 3 is a schematic perspective illustration also showing the end of the extruder **200** at the downstream side **210** and the honeycomb structure **201** (e.g., an extrudate) being extruded therefrom.

(25) The extruder **200** can be of any type, such as a ram extruder or a twin-screw extruder. For example, as shown in FIG. 2 the extruder **200** is a twin-screw extruder comprising a pair of

extruder screws **230** mounted in the barrel **202**. The first screw **230** and the second screw **232** may be arranged generally parallel to each other, as shown, although they may also be arranged at various angles relative to each other. The screws **230** can be coupled to a driving mechanism **236**, which can be located outside of the barrel **202**, for rotation in the same or different directions. Both the screws can be commonly coupled to a single driving mechanism **236** or, as shown, to individual driving mechanisms **236**. The screws **230**, ram, or other extrusion element moves the batch material **216** through the barrel **202** with pumping and/or mixing action in the axial direction **240**. As shown in FIGS. 2-3, an extruder assembly or cartridge located proximate the downstream end **210** can comprise extrusion hardware such as the honeycomb extrusion die **120**. A skin forming mask **205** can also be included for the honeycomb structure **201** with a peripheral outer skin. While extrusion is illustrated as being horizontally oriented in FIGS. 2 and 3, this disclosure is not so limited and extrusion can be horizontal, vertical, or at some incline thereto.

(26) When a desired length **316** is extruded, the green honeycomb structure **201** can be cut by any suitable means, such as via a wire, blade, saw, or the like to form a honeycomb body **203** as shown in FIG. 4. After cutting from the structure **201**, the honeycomb body **203** is in a green state and can be dried and fired via conventional methods to form a porous ceramic honeycomb body. The resulting porous ceramic body can have generally the same shape and configuration as the green honeycomb body, e.g., subject to shrinkage (if any) or other dimensional changes during drying, firing, or other manufacturing steps. Thus, the honeycomb body **203** in FIG. 4 generally represents the honeycomb body in both the green state (before firing) and the ceramic state (after firing).

(27) Upon exiting the extruder **200** in the axial direction **240**, the green honeycomb structure **201**, and therefore the honeycomb body **203** cut from the honeycomb structure **201**, comprises a honeycomb matrix **325** of axially extending and intersecting walls **312** that form a plurality of axially extending channels **308**. If the mask **205** is used, the honeycomb structure **201** can also comprise an axially extending outer peripheral surface **310** or the outer peripheral surface **310** can be applied to the honeycomb body **203** in a subsequent manufacturing step. The plurality of intersecting walls **312**, shown intersecting at perpendicular angles, form the channels **308** that extend in the axial direction **240**. For example, a representative channel **308R** extending in the axial direction **240** is shown by dashed lines for illustration purposes. A cross-sectional shape of the green honeycomb structure **201** perpendicular to the axial direction **240** can be circular (as shown), square, elliptical, rectangular, triangular, hexagonal, octagonal, or any other polygonal shape. Similarly, the channels **308** can have any suitable cross-sectional shape, such as square (as shown), circular, elliptical, rectangular, triangular, hexagonal, octagonal, or any other polygonal shape. The channels **308** can be all of the same shape and/or size, or different shapes and/or sizes. Similarly, the walls **312** can be all of the same thickness or different thicknesses (set by the width of the slots of the extrusion die **120**).

(28) Average cell density of the honeycomb body **203** when in the porous ceramic state after firing, can be any suitable value, such as between about 100 cells per square inch (cpsi) and about 900 cpsi (between about 15.5 cells per square cm and about 139.5 cells per square cm). The intersecting walls **312** can have any suitable transverse wall thickness, such as ranging from about 3.0 mils to 60 mils (about 0.076 mm to 1.52 mm). For example, the geometries of the porous ceramic honeycomb body may have an average cell density of 400 cpsi with a wall thickness of about 8 mils ("400/8") or with a wall thickness of about 6 mils ("400/6"). Other geometries of the porous ceramic honeycomb body can include, for example, combinations of (average cell density {in cpsi})/(wall thickness {in mil}) of 100/17, 200/12, 200/19, 270/19, 200/8, 300/8, 350/7, 600/4, 400/4, 600/3, 900/3, and the like.

(29) Accordingly, referring back to FIG. 1, FIG. 1 shows a partial, cross-sectional side view of a repeating structure of the honeycomb extrusion die **120** configured to form the intersecting walls **312**. This repeating die structure can be repeated throughout the die body **121** of the honeycomb extrusion die **120** and/or arranged with areas having different slot widths, pin sizes, or pins per

square inch.

(30) FIG. 5 shows an enlarged front view of a portion of the exit face **123** of the honeycomb extrusion die **120**. For example, the view of FIG. 5 may be provided at a central region of the exit face **123** of the honeycomb extrusion die **120** and not necessarily at an area proximate to the periphery of the exit face **123** that is configured to form the outer peripheral surface **310**. The inlet face **122** is located opposite the exit face **123**. Die body **121** can be made from any suitable tool material, such as tool steel. Portions of the honeycomb extrusion die **120**, such as the pin surfaces **128** defining the slots **130**, can be coated with a wear-resistant coating material. As described generally above, batch material **216** (e.g., see FIG. 2) enters the feedhole inlets **124A** (e.g., see FIG. 1) under pressure provided by the operation of the extruder **200** (e.g., see FIG. 2) and is discharged into the slots **130** at the feedhole outlets **124B**.

(31) The die pins **126** are arranged so that the side surfaces **128** (each pin **126** comprising four side surfaces **128** for square-shaped die pins **126** shown in FIG. 5) form the plurality of slots **130** defined between the respective opposing side surfaces **128** of the die pins **126**. The slots **130** intersect with the feedhole outlets **124B** of the feedholes **124** and extend axially to the exit face **123**. There may be overlap between the feedholes **124** and the slots **130**. During extrusion, the batch material **216** (e.g., see FIG. 2) flows through the slots **130** and forms the intersecting walls **312** (e.g., see FIGS. 3 and 4) when the batch material **216** is extruded through the honeycomb extrusion die **120**.

(32) In the embodiment depicted in FIGS. 1, 5 and 6, the die pins **126** are square-shaped when viewed from the exit face **123**. The square-shaped die pins **126** form square-shaped channels **308** (as correspondingly shown in FIGS. 3 and 4). In other embodiments, the die pins **126** can have shapes other than square, such as circular, triangular, rectangular non-square, hexagonal, octagonal, diamond, combinations of the aforementioned, and the like and they can comprise filleted or radiused corners when viewed in transverse cross-section, e.g., to minimize stress in the final porous ceramic honeycomb body and/or reduce puddling of any catalyst coating applied thereto.

(33) During use of the honeycomb extrusion die **120** in an extrusion process to form honeycomb green bodies, the batch material **216** is forced under pressure against the inlet face **122** and enters into the feedhole inlets **124A** of the feedholes **124**. The batch material **216** further flows in the feedholes **124** to the feedhole exits **124B** and transitions into the slots **130**. The structure of the slots **130** are defined by the plurality of die pins **126** extending to the exit face **123**, wherein the plurality of die pins **126** comprising side surfaces **128** configured to define the matrix of intersecting slots **130**.

(34) At least some of the slots **130** of the matrix of intersecting slots **130** are made up of entrance slot portions **135**, divots **132**, and exit slot portions **136**. In some embodiments, substantially all of the center region of the extrusion die **120**, except for the skin-forming region, comprises the above structure. In some embodiments, even the skin-forming region at the periphery of the extrusion die **120** comprises the entrance slot portions **135**, divots **132**, and exit slot portions **136**. In some embodiments, all of the matrix of intersecting slots **130** that define the walls **312** of the extruded honeycomb structure **201** and/or honeycomb body **203** comprise entrance slot portions **135**, divots **132**, and exit slot portions **136**. In some embodiments, all of the matrix of intersecting slots **130** (thus, including any slots in a skin forming region of the die) comprise the entrance slot portions **135**, divots **132**, and exit slot portions **136**.

(35) In more detail, the plurality of die pins **126** comprise divots **132** formed in the side surfaces **128** of the die pins **126** between the feedhole outlets **124B** and the exit face **123**. Divots **132**, as defined herein, are recessed portions at a transition between the entrance slot portions **135** and exit slot portions **136**. In some embodiments of the honeycomb extrusion die **120**, the divots **132** formed in the side surfaces **128** of the die pins **126** may have a divot width $D_{sub.D}$ of from 250 μm to 750 μm as measured from a centerline **135C** of the entrance slot portions **135** to the deepest part of the divot **132** in a direction perpendicular to the axial direction. Since the divot width $D_{sub.D}$ is

measured from the centerline **135C** of the slot portions **135**, the total or maximum width of the slots at the divots **132**, designated in FIG. **1** as divot slot width $W_{sub.D}$, is equal to twice the divot width $D_{sub.D}$. The divot slot width $W_{sub.D}$ is wider than both the entrance slot width $W_{sub.A}$ and the exit slot width $W_{sub.B}$. In some embodiments, the divots **132** have a divot length $L_{sub.D}$ of from 200 μm to 600 μm . Divots **132** can have a bottom configuration with a compound radius such as shown in FIG. **6**. Other honeycomb dies can have other divot configurations, such as the extrusion die **120B** having square-shaped divots **132B** of the sides **128** of die pins **126** shown in FIG. **7B**, such that the sides of the divot **132B** is flat.

(36) The entrance slot portions **135** of the matrix of intersecting slots **130** are positioned between intersections with the feedhole outlets **124B** and the divots **132**, whereas the exit slot portions **136** of the matrix of intersecting slots **130** are located between the divots **132** and the exit face **123**. As described above, the entrance slot portions **135** have the entrance slot width $W_{sub.A}$, and the exit slot portions **136** having the exit slot width $W_{sub.B}$. $W_{sub.A}$ and $W_{sub.B}$ can be measured as the dimensions of the slots as coated with a wear resistant coating.

(37) A relationship between $W_{sub.A}$ and $W_{sub.B}$ can be set to control or influence the high wear area of the honeycomb extrusion die **120** to occur at the exit slot portions **136**, irrespective of the batch material used. In particular, in some embodiments, the respective entrance slot portions **135**, and exit slot portions **136** are sized such that $W_{sub.A} > W_{sub.B}$ over the entire length of the slots **130**. That is, the entrance slot width $W_{sub.A}$ for any location along an entrance length $L_{sub.A}$ of the entrance slot portions **135** (from the intersections with the feed hole exits **124B** to the divots **132**, as shown in FIG. **6**) is greater than the exit slot width $W_{sub.B}$ for all locations along an exit length $L_{sub.B}$ of the exit slot portions **136** (from the divots **132** to the exit face **123**, as shown in FIG. **6**).

(38) According to some embodiments, the relationship between $W_{sub.A}$ and $W_{sub.B}$ is further set or defined with respect to a slot width contraction ratio (SWCR), where the slot width contraction ratio is defined as $SWCR = W_{sub.A} / W_{sub.B}$. In some embodiments, at least some of the matrix of intersecting slots **130** have a SWCR that is greater than or equal to 1.5. For example, the SWCR can be used to adjust the relative amounts of slip velocity of the batch material at a first location at the intersection of feedholes **124** and entrance slot portions **135**, and a second location at the exit slot portions **136**. In some embodiments of the honeycomb extrusion die **120**, the slot width contraction ratio (SWCR) is greater than 3.0. In some embodiments, SWCR values less than or equal to 1.05 may result in unwanted wear at the die pin root for some batch materials, while values of SWCR greater than 5.0 may cause excessive wear at the exit slot portions and thus premature wearing out of the honeycomb extrusion die. In some embodiments of the honeycomb extrusion die **120**, the SWCR can be from 2.0 to 5.0, such as from 3.0 to 5.0. Adjusting the relative amounts of slip velocity in the honeycomb extrusion die **120** is useful in die designs wherein the exit slot width $W_{sub.B}$ of individual slots is from 40 μm to 250 μm , although it may be useful on honeycomb extrusion dies with different exit slot widths.

(39) Regarding further dimensions of the slots **130** of the honeycomb extrusion die **120**, at least some of the entrance slot portions **135** have the entrance length $L_{sub.A}$ as measured between the intersections with the feed hole exits **124B** and the divots **132** (e.g., see FIG. **6**), and at least some of the exit slot portions **136** have the exit length $L_{sub.B}$ as measured between the divots **132** and the exit face **123** (e.g., see FIG. **6**), and wherein entrance length $L_{sub.A}$ is greater than the exit length $L_{sub.B}$. Having an entrance length $L_{sub.A}$ greater than the exit length $L_{sub.B}$ may assist in some embodiments to provide a die pressure reduction. In some embodiments, the entrance length $L_{sub.A}$ is greater than or equal to 1.1 times the exit length $L_{sub.B}$, or $L_{sub.A} \geq 1.1 L_{sub.B}$.

(40) Furthermore, the entrance length $L_{sub.A}$ of at least some of the entrance slot portions **135** in some embodiments is from 1270 μm to 2540 μm , as measured between the feedhole outlets **124B** and the divots **132** as shown in FIG. **6**. In some embodiments, at least some of the exit slot portions **136** have an exit length $L_{sub.B}$ from 508 μm to 1143 μm , as measured between the divots **132**

(downstream end thereof) and the exit face **123** as shown in FIG. **6**.

(41) In some embodiments, the entrance slot width $W_{sub.A}$ of the entrance slot portions **135** is a constant dimension along the entire length $L_{sub.A}$ of the entrance slot portions **135**. Similarly, in some embodiments, the exit slot width $W_{sub.B}$ of the exit slot portions **136** is a constant dimension along the entire length $L_{sub.B}$ of the exit slot portions **136**. However, the width of the slots **130** along the entrance slot portions **135** and/or the exit slot portions **136** can be non-constant for some or all of slots **130**.

(42) For example, in one alternative embodiment of the extrusion die **120**, designated as the extrusion die **120A** and shown in FIG. **7A**, the entrance slot width $W_{sub.A}$ of the entrance slot portion **135** is non-constant along its length $L_{sub.A}$ from the feedhole exits **124B** to the upstream side of the divot **132**. As such, the entrance slot portion **135** has tapered walls forming a wedge shape in plan view shown in FIG. **7A**. Others of the slots **130** can also have tapered walls and a wedge shaped configuration. A taper angle **137** measured between the walls as shown in FIG. **7A** can be from about 0.1 degrees to 10.0 degrees, for example. The other dimensions of $W_{sub.B}$ and $L_{sub.A}$, $L_{sub.B}$, and $L_{sub.D}$ and $D_{sub.D}$ can be the same as described above. In this embodiment, $W_{sub.A} > W_{sub.B}$ at each location along the length $L_{sub.A}$, and thus, even the minimum dimension of $W_{sub.A}$ is greater than the width $W_{sub.B}$ at any location along the exit slot portion **136**.

(43) The honeycomb extrusion die **120**, comprising the slot configurations with entrance slot portion **135**, divots **132**, and exit slot portions **136** can be manufactured in any suitable manner. In some embodiments, the die **120** is manufactured by a wire-electrical discharge machining (wire-EDM) technique and thereafter coated with a suitable wear coating to achieve the desired dimensions of $W_{sub.A}$ and $W_{sub.B}$.

(44) In another aspect, a method of manufacturing a honeycomb body (e.g., honeycomb body **203**) is disclosed. Reference is made to FIG. **8**, which shows a flowchart of the manufacturing method **800**. In particular, FIG. **8** describes a method of manufacturing a honeycomb structure (e.g., honeycomb structure **201**) using a honeycomb extrusion die (e.g., any of the honeycomb extrusion dies **120**, including the honeycomb extrusion dies **120A**, **120B** of FIGS. **7A** and **7B**).

(45) The method **800** comprises, in block **802**, providing a honeycomb extrusion die **120** comprising a die body (e.g., die body **121**) comprising an inlet face (e.g., inlet face **122**) and an exit face (e.g., exit face **123**), the die body having feedholes (e.g., feedholes **124**) with feedhole entrances (e.g., feedhole entrances **124A**) located at the inlet face and feedhole outlets (e.g., feedhole outlets **124B**), and a plurality of die pins (e.g., die pins **126**) extending a distance into the exit face, the plurality of die pins comprising side surfaces (e.g., side surfaces **128**) configured to define a matrix of intersecting slots (e.g., intersecting slots **130**), at least some of the matrix of intersecting slots and the plurality of die pins further comprising divots (e.g., divots **132**) formed in the side surfaces of the die pins between the feedholes and the exit face, entrance slot portions (e.g., entrance slot portions **135**) of the matrix of intersecting slots between the feedhole outlets and the divots, the entrance slot portions having an entrance slot width $W_{sub.A}$, and exit slot portions (e.g., exit slot portions **136**) of the matrix of intersecting slots between the divots and the exit face, the exit slot portions having an exit slot width $W_{sub.B}$, wherein $W_{sub.A} > W_{sub.B}$ over an entire length of the slots **130**.

(46) The method **800** further comprises, in block **804**, extruding a batch material (e.g., batch material **216**) through the feedholes and matrix of intersecting slots. According to the method **800**, a position of maximum die wear from the extruding of the batch material is located at the exit slot portions of the matrix of intersecting slots, and this occurs regardless of the rheology of the batch material, i.e., whether the batch material is a high die wear batch mixture or a low die wear batch mixture.

(47) It will be apparent to those skilled in the art that various modifications and variations can be made in the present disclosure without departing from the scope of the disclosure. Thus, it is

intended that the appended claims cover the modifications and variations of this disclosure provided they come within the scope of the appended claims and their equivalents.

Claims

1. A honeycomb extrusion die, comprising: a die body comprising: an inlet face and an exit face axially opposite from the inlet face; feedholes with feedhole entrances at the inlet face that terminate at feedhole outlets within the die body; and a plurality of die pins extending to the exit face, the plurality of die pins comprising side surfaces that define a matrix of intersecting slots, wherein at least some of the slots in the matrix of intersecting slots comprise: divots formed in the side surfaces of the die pins at positions located axially between the feedhole outlets and the exit face, entrance slot portions extending from intersections with the feedhole outlets to the divots, the entrance slot portions having an entrance slot width $W_{sub.A}$, and exit slot portions extending from the divots to the exit face, the exit slot portions having an exit slot width $W_{sub.B}$, wherein $W_{sub.A} > W_{sub.B}$ over an entire slot length from the feedhole outlets to the exit face.
2. The honeycomb extrusion die of claim 1, comprising a slot width contraction ratio SWCR over the entire slot length of the at least some of the slots of the matrix of intersecting slots is greater than or equal to 1.5, wherein the SWCR is defined as $SWCR = W_{sub.A} / W_{sub.B}$.
3. The honeycomb extrusion die of claim 2, wherein the SWCR is greater than or equal to 3.0.
4. The honeycomb extrusion die of claim 2, wherein the SWCR is from 1.5 to 5.0.
5. The honeycomb extrusion die of claim 4, wherein the SWCR is from 3.0 to 5.0.
6. The honeycomb extrusion die of claim 1, wherein the exit slot width $W_{sub.B}$ of the at least some of the slots of the matrix of intersecting slots is from 40 μm to 250 μm .
7. The honeycomb extrusion die of claim 1, wherein the entrance slot width $W_{sub.A}$ of at least some of the entrance slot portions is a constant dimension along an entire length of the entrance slot portions from the intersections with the feedhole outlets to the divots.
8. The honeycomb extrusion die of claim 1, wherein the exit slot width $W_{sub.B}$ of at least some of the exit slot portions is a constant dimension along an entire length of the exit slot portions from the divots to the exit face.
9. The honeycomb extrusion die of claim 1, wherein the divots formed in the side surfaces of the die pins have a divot width of from 250 μm to 750 μm measured from a centerline of the entrance slot portions.
10. The honeycomb extrusion die of claim 1, wherein the divots formed in the side surfaces of the die pins have a divot length $L_{sub.D}$ of from 200 μm to 600 μm .
11. The honeycomb extrusion die of claim 1, wherein the entrance slot portions have an entrance length $L_{sub.A}$ as measured between the intersections with the feedhole outlets and the divots, and the exit slot portions have an exit length $L_{sub.B}$ as measured between the divots and the exit face, and wherein $L_{sub.A} > L_{sub.B}$.
12. The honeycomb extrusion die of claim 11, wherein the entrance length $L_{sub.A}$ is from 1270 μm to 2540 μm .
13. The honeycomb extrusion die of claim 11, wherein the exit length $L_{sub.B}$ is from 508 μm to 1143 μm .
14. The honeycomb extrusion die of claim 11, wherein $L_{sub.A} \geq 1.1 L_{sub.B}$.
15. The honeycomb extrusion die of claim 14, wherein a slot width contraction ratio SWCR of the at least some of the slots of the matrix of intersecting slots is greater than or equal to 1.5, wherein the SWCR is defined as $SWCR = W_{sub.A} / W_{sub.B}$.
16. The honeycomb extrusion die of claim 1, wherein the entrance slot portions have an entrance length $L_{sub.A}$ from 1270 μm to 2540 μm as measured between the intersections with the feedhole outlets and the divots, the exit slot portions have an exit length $L_{sub.B}$ from 508 μm to 1143 μm as measured between the divots and the exit face, and the divots formed in the side surfaces of the die

pins have a divot width $D_{sub.D}$ from 250 μm to 750 μm measured from a centerline of the entrance slot portions, and a divot length $L_{sub.D}$ from 200 μm to 600 μm .

17. A method of manufacturing a honeycomb structure, comprising: extruding a batch material through the extrusion die of claim 1.

18. The method of manufacturing the honeycomb structure of claim 17, wherein a position of maximum die wear from the extruding is located at the exit slot portions of the slots of the matrix of intersecting slots.

19. The method manufacturing the honeycomb structure of claim 17, wherein the position of maximum die wear from the extruding is located at the exit slot portions of the slots of the matrix of intersecting slots regardless of a rheology of the batch material.

20. A method of fabricating a honeycomb extrusion die, comprising: creating feedholes in a die body, wherein the feedholes extend from feedhole entrances at an inlet face of the die body to feedhole outlets within the die body; creating a matrix of intersecting slots that extends axially into the die body from an exit face of the die body that is axially opposite to the inlet face, wherein the slots are defined by side surfaces of die pins extending to the exit face, and forming divots in the side surfaces of the die pins at positions located axially between the feedhole outlets and the exit face, wherein creating the matrix of intersecting slots comprises forming at least some of the slots with entrance slot portions having an entrance slot width $W_{sub.A}$ extending from intersections with the feedhole outlets to the divots, and with exit slot portions having an exit slot width $W_{sub.B}$ extending from the divots to the exit face, wherein $W_{sub.A} > W_{sub.B}$ over an entire slot length from the feedhole outlets to the exit face.
